Numerical Optimization Practice Set

Gradient Descent, Momentum, Adagrad, RMSprop, Adam

Deep Learning Optimization

Instructions

Perform **two iterations** ($t = 0 \to 1$, and $t = 1 \to 2$) for the following optimization problems.

- All calculations are shown in vectorized form where $\mathbf{w} = \begin{bmatrix} x \\ y \end{bmatrix}$.
- Unless specified otherwise, initialize accumulators (velocity $\mathbf{v}$, squared grad $\mathbf{G}$, moments $\mathbf{m}, \mathbf{v}$) to $\mathbf{0}$.
- Assume smoothing term $\epsilon \approx 0$ unless specified.

Contents

1 Gradient Descent (GD)

Update Rule: $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t)$

1.1 Level 1: Easy (Isotropic)

Problem: Minimize $f(x, y) = 0.5x^2 + 0.5y^2$.

Given: $\mathbf{w}_0 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, Learning Rate $\eta = 0.1$.

Step-by-Step Solution

Gradient Formula: $\nabla f(\mathbf{w}) = \begin{bmatrix} x \\ y \end{bmatrix}$.

Iteration 1 ($t = 0$):

$$\begin{aligned} \text{Gradient:} \quad \nabla f(\mathbf{w}_0) &= \begin{bmatrix} 4 \\ 2 \end{bmatrix} \\ \text{Update:} \quad \mathbf{w}_1 &= \begin{bmatrix} 4 \\ 2 \end{bmatrix} - 0.1 \begin{bmatrix} 4 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 - 0.4 \\ 2 - 0.2 \end{bmatrix} = \begin{bmatrix} 3.6 \\ 1.8 \end{bmatrix} \end{aligned}$$

Iteration 2 ($t = 1$):

$$\begin{aligned} \text{Gradient:} \quad \nabla f(\mathbf{w}_1) &= \begin{bmatrix} 3.6 \\ 1.8 \end{bmatrix} \\ \text{Update:} \quad \mathbf{w}_2 &= \begin{bmatrix} 3.6 \\ 1.8 \end{bmatrix} - 0.1 \begin{bmatrix} 3.6 \\ 1.8 \end{bmatrix} \\ &= \begin{bmatrix} 3.6 - 0.36 \\ 1.8 - 0.18 \end{bmatrix} = \begin{bmatrix} 3.24 \\ 1.62 \end{bmatrix} \end{aligned}$$

1.2 Level 2: Medium (Anisotropic)

Problem: Minimize $f(x, y) = 2x^2 + y^2$.

Given: $\mathbf{w}_0 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, $\eta = 0.1$.

Step-by-Step Solution

Gradient Formula: $\nabla f(\mathbf{w}) = \begin{bmatrix} 4x \\ 2y \end{bmatrix}$.

Iteration 1 ($t = 0$):

$$\nabla f(\mathbf{w}_0) = \begin{bmatrix} 4(2) \\ 2(5) \end{bmatrix} = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix} - 0.1 \begin{bmatrix} 8 \\ 10 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 4.0 \end{bmatrix}$$

Iteration 2 ($t = 1$):

$$\nabla f(\mathbf{w}_1) = \begin{bmatrix} 4(1.2) \\ 2(4.0) \end{bmatrix} = \begin{bmatrix} 4.8 \\ 8.0 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} 1.2 \\ 4.0 \end{bmatrix} - 0.1 \begin{bmatrix} 4.8 \\ 8.0 \end{bmatrix} = \begin{bmatrix} 1.2 - 0.48 \\ 4.0 - 0.8 \end{bmatrix} = \begin{bmatrix} 0.72 \\ 3.20 \end{bmatrix}$$

1.3 Level 3: Difficult (Coupled Variables)

Problem: Minimize $f(x, y) = x^2 + xy + y^2$.
Given: $\mathbf{w}_0 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $\eta = 0.2$.

Step-by-Step Solution

Gradient Formula: $\nabla f(\mathbf{w}) = \begin{bmatrix} 2x + y \\ x + 2y \end{bmatrix}$.

Iteration 1 ($t = 0$):

$$\nabla f(\mathbf{w}_0) = \begin{bmatrix} 2(1) + (-2) \\ 1 + 2(-2) \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} - 0.2 \begin{bmatrix} 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 + 0.6 \end{bmatrix} = \begin{bmatrix} 1 \\ -1.4 \end{bmatrix}$$

Iteration 2 ($t = 1$):

$$\nabla f(\mathbf{w}_1) = \begin{bmatrix} 2(1) + (-1.4) \\ 1 + 2(-1.4) \end{bmatrix} = \begin{bmatrix} 0.6 \\ -1.8 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} 1 \\ -1.4 \end{bmatrix} - 0.2 \begin{bmatrix} 0.6 \\ -1.8 \end{bmatrix} = \begin{bmatrix} 1 - 0.12 \\ -1.4 + 0.36 \end{bmatrix} = \begin{bmatrix} 0.88 \\ -1.04 \end{bmatrix}$$

2 Momentum Based GD

Update Rules: 1. $\mathbf{v}_{t+1} = \gamma \mathbf{v}_t + \eta \nabla f(\mathbf{w}_t)$ 2. $\mathbf{w}_{t+1} = \mathbf{w}_t - \mathbf{v}_{t+1}$

2.1 Level 1: Easy

Problem: $f(x, y) = x^2 + 2y^2$. $\mathbf{w}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \eta = 0.1, \gamma = 0.9$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 2x \\ 4y \end{bmatrix}$.

Iteration 1 ($t = 0$):

$$\nabla f_0 = \begin{bmatrix} 2(1) \\ 4(1) \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$\mathbf{v}_1 = 0.9 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 0.1 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$$

Iteration 2 ($t = 1$):

$$\nabla f_1 = \begin{bmatrix} 2(0.8) \\ 4(0.6) \end{bmatrix} = \begin{bmatrix} 1.6 \\ 2.4 \end{bmatrix}$$

$$\mathbf{v}_2 = 0.9 \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix} + 0.1 \begin{bmatrix} 1.6 \\ 2.4 \end{bmatrix} = \begin{bmatrix} 0.18 + 0.16 \\ 0.36 + 0.24 \end{bmatrix} = \begin{bmatrix} 0.34 \\ 0.60 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} - \begin{bmatrix} 0.34 \\ 0.60 \end{bmatrix} = \begin{bmatrix} 0.46 \\ 0.00 \end{bmatrix}$$

2.2 Level 2: Medium (Overshoot Risk)

Problem: $f(x, y) = 5x^2 + y$. $\mathbf{w}_0 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \eta = 0.1, \gamma = 0.5$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 10x \\ 1 \end{bmatrix}$.

Iteration 1 ($t = 0$)**:**

$$\begin{aligned} \nabla f_0 &= \begin{bmatrix} 20 \\ 1 \end{bmatrix} \\ \mathbf{v}_1 &= 0.5 \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 0.1 \begin{bmatrix} 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0.1 \end{bmatrix} \\ \mathbf{w}_1 &= \begin{bmatrix} 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.1 \end{bmatrix} \end{aligned}$$

Iteration 2 ($t = 1$)**:**

$$\begin{aligned} \nabla f_1 &= \begin{bmatrix} 10(0) \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \mathbf{v}_2 &= 0.5 \begin{bmatrix} 2 \\ 0.1 \end{bmatrix} + 0.1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.05 + 0.1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.15 \end{bmatrix} \\ \mathbf{w}_2 &= \begin{bmatrix} 0 \\ -0.1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.15 \end{bmatrix} = \begin{bmatrix} -1 \\ -0.25 \end{bmatrix} \end{aligned}$$

2.3 Level 3: Difficult (Saddle Point)

Problem: $f(x, y) = x^2 - y^2$. $\quad \mathbf{w}_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \eta = 0.2, \gamma = 0.8.$

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 2x \\ -2y \end{bmatrix}$.

Iteration 1 ($t = 0$)**:**

$$\begin{aligned}
\nabla f_0 &= \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\
\mathbf{v}_1 &= 0.2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} \\
\mathbf{w}_1 &= \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}
\end{aligned}$$

Iteration 2 ($t = 1$)**:**

$$\begin{aligned}
\nabla f_1 &= \begin{bmatrix} 0.6 \\ -1.4 \end{bmatrix} \\
\mathbf{v}_2 &= 0.8 \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} + 0.2 \begin{bmatrix} 0.6 \\ -1.4 \end{bmatrix} = \begin{bmatrix} 0.16 + 0.12 \\ -0.16 - 0.28 \end{bmatrix} = \begin{bmatrix} 0.28 \\ -0.44 \end{bmatrix} \\
\mathbf{w}_2 &= \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix} - \begin{bmatrix} 0.28 \\ -0.44 \end{bmatrix} = \begin{bmatrix} 0.02 \\ 1.14 \end{bmatrix}
\end{aligned}$$

3 Adagrad

Update Rules: 1. $\mathbf{G}_t = \mathbf{G}_{t-1} + (\nabla f_t)^2$ 2. $\mathbf{w}_{t+1} = \mathbf{w}_t - \frac{\eta}{\sqrt{\mathbf{G}_t + \epsilon}} \odot \nabla f_t$

3.1 Level 1: Easy

Problem: $f(x, y) = 2x^2 + 2y^2$. $\mathbf{w}_0 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \eta = 0.5$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 4x \\ 4y \end{bmatrix}$.

Iteration 1 **($t = 0$):**

$$\nabla f_0 = \begin{bmatrix} 4 \\ 12 \end{bmatrix} \implies (\nabla f_0)^2 = \begin{bmatrix} 16 \\ 144 \end{bmatrix}$$

$$\mathbf{G}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 16 \\ 144 \end{bmatrix} = \begin{bmatrix} 16 \\ 144 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} - 0.5 \begin{bmatrix} \frac{4}{\sqrt{16}} \\ \frac{12}{\sqrt{144}} \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} - 0.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 2.5 \end{bmatrix}$$

Iteration 2 **($t = 1$):**

$$\nabla f_1 = \begin{bmatrix} 2 \\ 10 \end{bmatrix} \implies (\nabla f_1)^2 = \begin{bmatrix} 4 \\ 100 \end{bmatrix}$$

$$\mathbf{G}_1 = \begin{bmatrix} 16 \\ 144 \end{bmatrix} + \begin{bmatrix} 4 \\ 100 \end{bmatrix} = \begin{bmatrix} 20 \\ 244 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.5 \\ 2.5 \end{bmatrix} - 0.5 \begin{bmatrix} \frac{2}{\sqrt{20}} \\ \frac{10}{\sqrt{244}} \end{bmatrix} \approx \begin{bmatrix} 0.5 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 0.223 \\ 0.320 \end{bmatrix} = \begin{bmatrix} 0.277 \\ 2.180 \end{bmatrix}$$

3.2 Level 2: Medium

Problem: $f(x, y) = 0.5x^2 + y$. $\mathbf{w}_0 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \eta = 1.0$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} x \\ 1 \end{bmatrix}$.

Iteration 1 $(t = 0)$**:**

$$\nabla f_0 = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \implies \mathbf{G}_0 = \begin{bmatrix} 16 \\ 1 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 4 \\ 0 \end{bmatrix} - 1.0 \begin{bmatrix} 4/\sqrt{16} \\ 1/\sqrt{1} \end{bmatrix} = \begin{bmatrix} 4-1 \\ 0-1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Iteration 2 $(t = 1)$**:**

$$\nabla f_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \implies (\nabla f_1)^2 = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$$
$$\mathbf{G}_1 = \begin{bmatrix} 16 \\ 1 \end{bmatrix} + \begin{bmatrix} 9 \\ 1 \end{bmatrix} = \begin{bmatrix} 25 \\ 2 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} - 1.0 \begin{bmatrix} 3/\sqrt{25} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 3-0.6 \\ -1-0.707 \end{bmatrix} = \begin{bmatrix} 2.400 \\ -1.707 \end{bmatrix}$$

3.3 Level 3: Difficult (Sparse Feature Simulation)

Problem: $f(x, y) = 10x^2 + y^2$. $\mathbf{w}_0 = \begin{bmatrix} 1 \\ 10 \end{bmatrix}, \eta = 0.1$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 20x \\ 2y \end{bmatrix}$.

Iteration 1 $(t = 0)$**:**

$$\nabla f_0 = \begin{bmatrix} 20 \\ 20 \end{bmatrix} \implies \mathbf{G}_0 = \begin{bmatrix} 400 \\ 400 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 10 \end{bmatrix} - 0.1 \begin{bmatrix} 20/\sqrt{400} \\ 20/\sqrt{400} \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix} - \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.9 \\ 9.9 \end{bmatrix}$$

Iteration 2 $(t = 1)$**:**

$$\nabla f_1 = \begin{bmatrix} 18 \\ 19.8 \end{bmatrix} \implies (\nabla f_1)^2 = \begin{bmatrix} 324 \\ 392.04 \end{bmatrix}$$
$$\mathbf{G}_1 = \begin{bmatrix} 400 \\ 400 \end{bmatrix} + \begin{bmatrix} 324 \\ 392.04 \end{bmatrix} = \begin{bmatrix} 724 \\ 792.04 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} 0.9 \\ 9.9 \end{bmatrix} - 0.1 \begin{bmatrix} 18/\sqrt{724} \\ 19.8/\sqrt{792.04} \end{bmatrix}$$
$$\approx \begin{bmatrix} 0.9 \\ 9.9 \end{bmatrix} - \begin{bmatrix} 0.067 \\ 0.070 \end{bmatrix} = \begin{bmatrix} 0.833 \\ 9.830 \end{bmatrix}$$

4 RMSprop

Update Rules: 1. $E[g^2]_t = \beta E[g^2]_{t-1} + (1-\beta)(\nabla f_t)^2$ 2. $\mathbf{w}_{t+1} = \mathbf{w}_t - \frac{\eta}{\sqrt{E[g^2]_t}} \odot \nabla f_t$
(Using $\beta = 0.9$)

4.1 Level 1: Easy

Problem: $f(x,y) = x^2 + y^2$. $\mathbf{w}_0 = \begin{bmatrix}10\\10\end{bmatrix}, \eta = 0.1$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix}2x\\2y\end{bmatrix}$.

Iteration 1 ($t=0$):

$$\nabla f_0 = \begin{bmatrix}20\\20\end{bmatrix} \implies (\nabla f)^2 = \begin{bmatrix}400\\400\end{bmatrix}$$
$$E[g^2]_0 = 0.9(\mathbf{0}) + 0.1\begin{bmatrix}400\\400\end{bmatrix} = \begin{bmatrix}40\\40\end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix}10\\10\end{bmatrix} - 0.1\begin{bmatrix}20/\sqrt{40}\\20/\sqrt{40}\end{bmatrix} = \begin{bmatrix}10\\10\end{bmatrix} - \begin{bmatrix}0.316\\0.316\end{bmatrix} = \begin{bmatrix}9.684\\9.684\end{bmatrix}$$

Iteration 2 ($t=1$):

$$\nabla f_1 = \begin{bmatrix}19.368\\19.368\end{bmatrix} \implies (\nabla f)^2 \approx \begin{bmatrix}375.12\\375.12\end{bmatrix}$$
$$E[g^2]_1 = 0.9\begin{bmatrix}40\\40\end{bmatrix} + 0.1\begin{bmatrix}375.12\\375.12\end{bmatrix} = \begin{bmatrix}73.51\\73.51\end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix}9.684\\9.684\end{bmatrix} - 0.1\begin{bmatrix}19.368/\sqrt{73.51}\end{bmatrix} \approx \begin{bmatrix}9.458\\9.458\end{bmatrix}$$

4.2 Level 2: Medium

Problem: $f(x,y) = 2x^2 - 3y$. $\mathbf{w}_0 = \begin{bmatrix}1\\1\end{bmatrix}, \eta = 0.1$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 4x \\ -3 \end{bmatrix}$.

Iteration 1 $(t = 0)$**:**

$$\nabla f_0 = \begin{bmatrix} 4 \\ -3 \end{bmatrix} \implies (\nabla f)^2 = \begin{bmatrix} 16 \\ 9 \end{bmatrix}$$
$$E[g^2]_0 = 0.1 \begin{bmatrix} 16 \\ 9 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 0.9 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 4/\sqrt{1.6} \\ -3/\sqrt{0.9} \end{bmatrix} = \begin{bmatrix} 1 - 0.316 \\ 1 - (-0.316) \end{bmatrix} = \begin{bmatrix} 0.684 \\ 1.316 \end{bmatrix}$$

Iteration 2 $(t = 1)$**:**

$$\nabla f_1 = \begin{bmatrix} 2.736 \\ -3 \end{bmatrix} \implies (\nabla f)^2 = \begin{bmatrix} 7.485 \\ 9 \end{bmatrix}$$
$$E[g^2]_1 = 0.9 \begin{bmatrix} 1.6 \\ 0.9 \end{bmatrix} + 0.1 \begin{bmatrix} 7.485 \\ 9 \end{bmatrix} = \begin{bmatrix} 1.44 + 0.748 \\ 0.81 + 0.9 \end{bmatrix} = \begin{bmatrix} 2.188 \\ 1.710 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} 0.684 \\ 1.316 \end{bmatrix} - 0.1 \begin{bmatrix} 2.736/1.479 \\ -3/1.307 \end{bmatrix} = \begin{bmatrix} 0.499 \\ 1.545 \end{bmatrix}$$

4.3 Level 3: Difficult (Stagnation Check)

Problem: $f(x, y) = x^2 + 10y^2$. $\quad \mathbf{w}_0 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \eta = 0.1$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 2x \\ 20y \end{bmatrix}$.

Iteration 1 **($t = 0$):**

$$\nabla f_0 = \begin{bmatrix} 4 \\ 40 \end{bmatrix} \implies (\nabla f)^2 = \begin{bmatrix} 16 \\ 1600 \end{bmatrix}$$
$$E[g^2]_0 = 0.1 \begin{bmatrix} 16 \\ 1600 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 160 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - 0.1 \begin{bmatrix} 4/\sqrt{1.6} \\ 40/\sqrt{160} \end{bmatrix} = \begin{bmatrix} 2 - 0.316 \\ 2 - 0.316 \end{bmatrix} = \begin{bmatrix} 1.684 \\ 1.684 \end{bmatrix}$$

Iteration 2 **($t = 1$):**

$$\nabla f_1 = \begin{bmatrix} 3.368 \\ 33.68 \end{bmatrix} \implies (\nabla f)^2 \approx \begin{bmatrix} 11.34 \\ 1134 \end{bmatrix}$$
$$E[g^2]_1 = 0.9 \begin{bmatrix} 1.6 \\ 160 \end{bmatrix} + 0.1 \begin{bmatrix} 11.34 \\ 1134 \end{bmatrix} = \begin{bmatrix} 2.574 \\ 257.4 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} 1.684 \\ 1.684 \end{bmatrix} - 0.1 \begin{bmatrix} 3.368/1.604 \\ 33.68/16.04 \end{bmatrix} = \begin{bmatrix} 1.474 \\ 1.474 \end{bmatrix}$$

5 Adam (Adaptive Moment Estimation)

Update Rules: $\beta_1 = 0.9, \beta_2 = 0.999$. 1. $\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\nabla f_t$ 2. $\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)(\nabla f_t)^2$
3. Bias Correct: $\hat{\mathbf{m}}_t = \mathbf{m}_t/(1-\beta_1^{t+1})$, $\hat{\mathbf{v}}_t = \mathbf{v}_t/(1-\beta_2^{t+1})$
4. Update: $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t}}$

5.1 Level 1: Easy

Problem: $f(x, y) = x$. $\mathbf{w}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \eta = 0.1$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Iteration 1 ($t = 0$):

$$\mathbf{m}_0 = 0.1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_0 = 0.001 \begin{bmatrix} 1 \\ 0 \end{bmatrix}^2 = \begin{bmatrix} 0.001 \\ 0 \end{bmatrix}$$
$$\hat{\mathbf{m}}_0 = \frac{\mathbf{m}_0}{0.1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{v}}_0 = \frac{\mathbf{v}_0}{0.001} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 1/1 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 0 \end{bmatrix}$$

Iteration 2 ($t = 1$):

$$\mathbf{m}_1 = 0.9 \begin{bmatrix} 0.1 \\ 0 \end{bmatrix} + 0.1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.19 \\ 0 \end{bmatrix}$$
$$\mathbf{v}_1 = 0.999 \begin{bmatrix} 0.001 \\ 0 \end{bmatrix} + 0.001 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.001999 \\ 0 \end{bmatrix}$$
$$\hat{\mathbf{m}}_1 = \frac{\mathbf{m}_1}{1-0.81} = \frac{\mathbf{m}_1}{0.19} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{v}}_1 = \frac{\mathbf{v}_1}{1-0.998} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$\mathbf{w}_2 = \begin{bmatrix} -0.1 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.2 \\ 0 \end{bmatrix}$$

5.2 Level 2: Medium (Quadratic)

Problem: $f(x, y) = 2x^2 + 2y^2$. $\mathbf{w}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \eta = 0.01$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 4x \\ 4y \end{bmatrix}$.

Iteration 1 ($t = 0$)**:**

$$\nabla f_0 = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\mathbf{m}_0 = 0.1 \begin{bmatrix} 4 \\ -4 \end{bmatrix} = \begin{bmatrix} 0.4 \\ -0.4 \end{bmatrix}, \quad \mathbf{v}_0 = 0.001 \begin{bmatrix} 16 \\ 16 \end{bmatrix} = \begin{bmatrix} 0.016 \\ 0.016 \end{bmatrix}$$

$$\hat{\mathbf{m}}_0 = \mathbf{m}_0/0.1 = \begin{bmatrix} 4 \\ -4 \end{bmatrix}, \quad \hat{\mathbf{v}}_0 = \mathbf{v}_0/0.001 = \begin{bmatrix} 16 \\ 16 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} - 0.01 \begin{bmatrix} 4/\sqrt{16} \\ -4/\sqrt{16} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0.01 \\ -0.01 \end{bmatrix} = \begin{bmatrix} 0.99 \\ -0.99 \end{bmatrix}$$

5.3 Level 3: Difficult (Normalization Effect)

Problem: $f(x, y) = 0.1x^2 + 0.1y^2$. $\quad \mathbf{w}_0 = \begin{bmatrix} 10 \\ 10 \end{bmatrix}, \eta = 0.1$.

Step-by-Step Solution

Gradient: $\nabla f = \begin{bmatrix} 0.2x \\ 0.2y \end{bmatrix}$.

Iteration 1 ($t = 0$)**:**

$$\nabla f_0 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$\mathbf{m}_0 = \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix}, \quad \mathbf{v}_0 = \begin{bmatrix} 0.004 \\ 0.004 \end{bmatrix}$$

$$\hat{\mathbf{m}}_0 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \hat{\mathbf{v}}_0 = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$\mathbf{w}_1 = \begin{bmatrix} 10 \\ 10 \end{bmatrix} - 0.1 \begin{bmatrix} 2/2 \\ 2/2 \end{bmatrix} = \begin{bmatrix} 9.9 \\ 9.9 \end{bmatrix}$$

Note: Despite small gradients, Adam normalizes the step size magnitude to η.